



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2023 – 0014



Agricultural Machinery – Biomass Shredder RUMVD 4500-14-Y

Test Report Number	2023 – 0014
Date of Issuance	February 06, 2023
Valid Until	February 06, 2028
Agricultural Machinery	Biomass Shredder
Type	Horizontal, Rotating blade, Chipping and shredding, Engine-driven, Mobile (Cart type)
Brand	RUMVD
Model	4500-14-Y
Test Requested by	RU FOUNDRY & MACHINE SHOP CORPORATION Sitio Aning, Brgy. Pahanocoy, Bacolod City

SUMMARY

Performance Criteria	Requirement as per PNS/PAES 244:2010	AMTEC Test
Input Capacity, kg/h	1000 ^a	2073.71
Shredding Efficiency, %, minimum	90	91.50
Noise Level, dB(A), maximum	96	102.3

^a As specified by the manufacturer.

OBSERVATIONS

1. Operator's manual (Based on PAES102:2000)	
a. Language	English and Filipino
b. Safety and warnings	Not provided
c. Operating information	Provided
d. Accessories and attachments	Not provided
e. Maintenance instruction	Provided
f. Storage	Not provided
g. Handling, reception, transportation, assembly, and installation	Not provided
h. Specifications	Not provided
i. Dismantling and disposal	Not provided
j. Warranty	Provided
k. Parts list	Provided
2. Previous test reports of similar brand and model	2021-0259
3. Manufacturer	RU FOUNDRY & MACHINE SHOP CORPORATION
4. Prime mover used	5.22 kW YANMAR TF70L Diesel Engine
5. Number of operators	Three (One feeding operator and two feeding assistants)
6. Loading of input	Input materials were manually loaded into the hopper. The feeding operator used a wooden stick to push the input materials further into the shredding chamber.
7. Adjusting and repairing of parts	Locally fabricated parts can be repaired. Belt tension can be adjusted through the adjustable engine base.
8. Collecting of output	The output materials fell on the ground and were collected after the testing operation.
9. Cleaning of parts	Parts are accessible for cleaning.
10. Transporting the machine	Wheels and tow bar are provided for transport.
11. Safety features	1. A safety cover for the transmission system was provided. 2. The feeding operator is equipped with an ear protection device, rubber gloves, and face shield during the test operation.
12. Failure or abnormalities that may be observed on the machine or its component parts during and after the operation	No breakdown occurred during the test.
13. Steel bars, metal sheet or plate and heavy-duty mild steel shall be generally used for the different components of the machine	Checkered plates, angle bars, and metal plates were used for the different components of the machine.

OBSERVATIONS (Cont'n)

14. Provided with output chute	Provided
15. Knives or cutting mechanisms shall be made of high carbon steel and/or tool steel	The knives are made up of AISI 4140 / AISI 4142 low alloy steel as per the MIRDC certification submitted by the requesting party.
16. The angle of inclination with respect to horizontal axis for blades or cutting mechanism attached to the cylinder (For open cylinder only).	Not verified
17. Clearance between rotating blades and counter blades of the shredding chamber must be at least 6 mm.	Not applicable
18. Cover for shredding blades must be at least 2.5 mm M.S. plate and/or B.I. plate or its equivalent material	The cover for shredding blades used is made of cast iron and has a thickness of 19.30 mm.
19. Bolts and screws shall conform to the requirements of PAES 311 and 313	Not verified
20. Sizes of the parts of the machine shall be based on the specifications of the manufacturer	Refer to specification's table on page 6.
21. The base shall be rigid and its rotating components shall be statically and dynamically balanced for stable operation	The base is made of welded angle bars. However, it was not verified if it was statically and dynamically balanced.
22. Machine shall be free from manufacturing defects	Based on ocular inspections, no manufacturing defects were observed.
23. Cutting mechanism shall be adjustable, replaceable and can be sharpened	Blades can be replaced and sharpened.
24. All surfaces shall be free from rust and shall be coated with a suitable paint material	Non-rotating parts were painted.
25. Belt cover or guard and provisions for belt tightening and adjustments shall be provided	Belt guard is provided. However, there is no provision for belt tightening and adjustments.
26. Mechanism for intermediate load disengagement between prime mover and biomass shredder shall be provided	Not provided
27. All welded parts shall be water-tight and smoothly polished and it shall pass visual inspection criteria for discontinuity of materials	Not verified
28. Welded joints shall not be less than 4 mm side filler welded. Undercut shall not exceed 2mm for any length of weld	Not verified
29. Other remarks	The machine can also be used as a wood chipper. Only the shredding operation was tested.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date of Test	January 19, 2023
Location of Test	Sitio Aning, Brgy. Pahanocoy, Bacolod City 10°36'46.9" N, 122°55'27.9" E
Items Tested	Operator's Manual, Shredding Efficiency, Input Capacity, Speed of Components, Noise Level, Fuel Consumption Rate, and Number of Operators

METHODS OF TEST

The biomass shredder was tested based on the PNS/PAES 245:2010 Agricultural Machinery – Biomass Shredder – Methods of Test.

DESCRIPTION OF THE MACHINE

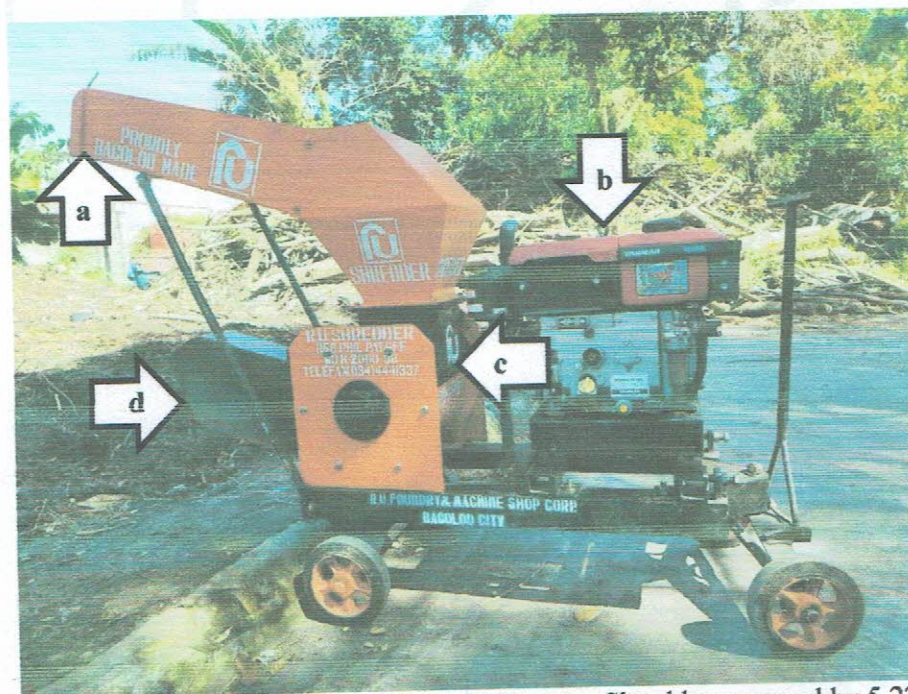


Figure 1. Parts of the RUMVD 4500-14-Y Biomass Shredder powered by 5.22 kW YANMAR TF70L Diesel Engine include the following: (a) Input chute, (b) Prime mover, (c) Shredding chamber, and (d) Outlet.

SPECIFICATIONS

Item		Manufacturer's Specification ^a	AMTEC Verification
1	Main structure		
1.1	Overall dimensions, mm		
1.1.1	Length	1840	1905
1.1.2	Width	660	650
1.1.3	Height	1258	1295
1.2	Weight without primemover, kg	ND	NM
2	Input chute		
2.1	Material	Checkered plate	Checkered plate
2.2	Bottom opening, mm		
2.2.1	Length	340	180
2.2.2	Width	203	180
2.3	Height from the ground, mm	1115	1135
3	Shredder assembly		
3.1	Blade cover		
3.1.1	Type	Cylindrical	Cylindrical
3.1.2	Materials	Cast iron	Cast Iron
3.2	Blade		
3.2.1	Material	AISI 4140/ AISI 4142 (Low alloy steel) ^b	AISI 4140/ AISI 4142 (Low alloy steel) ^b
3.2.2	Dimensions, mm		
3.2.2.1	Length	241	122.5
3.2.2.2	Width	63	74.74
3.2.2.3	Thickness	7	5.55
3.2.2.4	Shape	ND	Rectangular with curved sharpened side
3.2.3	Number of blades	9	18
3.2.4	Clearance to the shredding cylinder, mm	ND	6.45

^a Data were obtained from the specifications sheet provided by the requesting party.

^b The data were based on MIRDC certification submitted by the requesting party.

Note: ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification ^a	AMTEC Verification
4	Output chute		
4.1	Material	Checkered plate	Checkered plate
4.2	Dimensions, mm		
4.2.1	Length	230	225
4.2.2	Width	118	115
4.3	Height from the ground, mm	370	400
5	Prime mover		
5.1	Engine		
5.1.1	Brand	YANMAR	YANMAR
5.1.2	Model	TF70L	TF70L
5.1.3	Make or manufacturer	YANMAR CO., LTD, INDONESIA	YANMAR CO., LTD, INDONESIA
5.1.4	Serial number	TF70L-1S AO32CI	TF70L-1S AO32CI
5.1.5	Type	ND	Four-stroke cycle, single horizontal cylinder
5.1.6	Rated power, kW	5.22	NM
5.1.7	Rated speed, rpm	2400	NM
5.1.8	Displacement, cm ³	ND	NM
5.1.9	Fuel type	ND	Diesel
5.1.10	Fuel tank capacity, L	ND	NM
5.1.11	Cooling system	ND	Water-cooled
5.1.12	Starting system	ND	Manual cranking lever
6	Power transmission system		
6.1	Engine to shredding cylinder	Belt and pulley	V-belt and pulley transmission system
6.2	Engine ^b	ND	151 × 2 × 32
6.3	Shredding cylinder ^b	ND	76 × 2 × 24
6.4	Belt size	B-59	A-59
7	Safety devices	ND	Belt and pulley transmission system is provided with cover.
8	Special features	ND	Can also be used as wood chipper

^a Data were obtained from the engine nameplate and specifications sheet provided by the requesting party.

^b Pulley diameter, mm × number of belts × shaft diameter, mm

Note: ND – No Data, NM – Not Measured

RESULTS

Item		Data	
1	Condition of test material		
1.1	Test material	Chopped young coconut	
1.2	Variety	ND	
1.3	Source	Bacolod City	
1.4	Moisture content, % _{w.b.}	86.20	
1.5	Dimensions, mm		
1.5.1	Length	119.2	
1.5.2	Width/diameter	69.7	
1.5.3	Thickness	36.09	
2	Performance test		
2.1	Weight of input, kg	341.6	
2.2	Input capacity, kg/h	2073.7	
2.3	Shredding efficiency, %	91.54	
2.4	Unshredded, %	8.46	
2.5	Speed of components, rpm	Without load	With load
2.5.1	Engine shaft	2259	2157
2.5.2	Shredding shaft	4614	4298
2.6	Noise level, dB(A)	Without load	With load
2.6.1	Feeding operator	96.5	102.3
2.6.2	Feeding assistant	96.5	98.1
2.7	Fuel consumption rate, L/h	1.54	

Note: ND – No Data

OBSERVATIONS



Figure 2. Quality of output

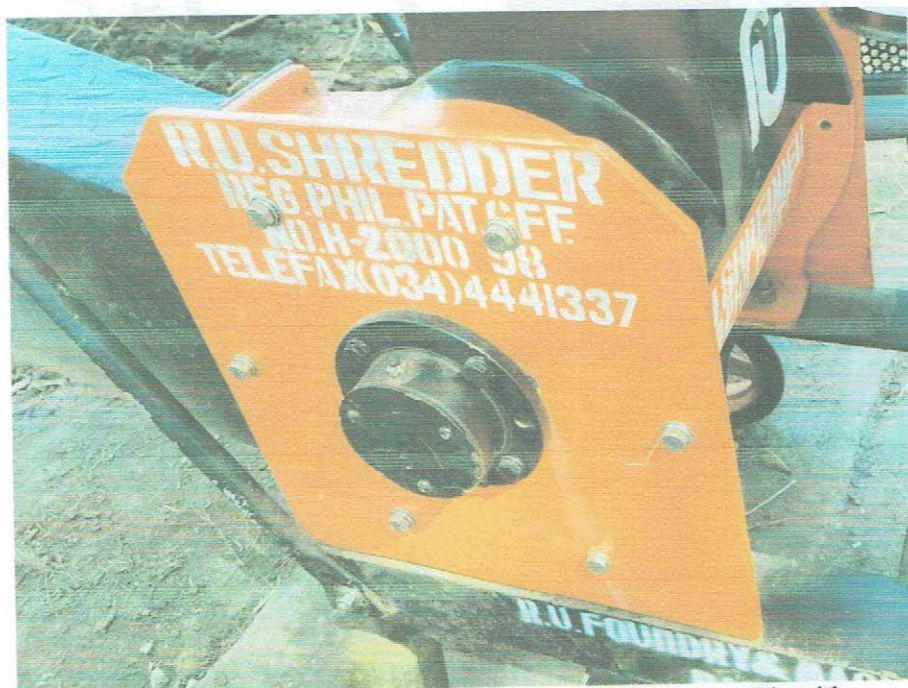


Figure 3. Marking on the RUMVD 4500-14-Y Biomass Shredder

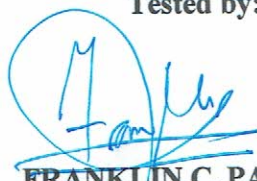
OBSERVATIONS (Cont'n)

Figure 4. Markings on the RUMVD 4500-14-Y Biomass Shredder

OBSERVATIONS (Cont'n)

Figure 5. The feeding operator is provided with ear protection device, cotton gloves, and face shield during the test operation.

Tested by:

**FRANKLIN C. PAJARON**

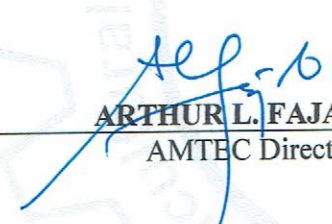
Test Engineer

Verified by:


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Test Engineer

Approved for Release:


ARTHUR L. FAJARDO

AMTEC Director

MAY 12 2023

Date

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